

James Qin

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LinkedIn: linkedin.com/in/jamesqin | **GitHub:** github.com/Jamchin | **Portfolio:** jamchin.github.io

EDUCATION

Northeastern University, Boston, MA

Expected May 2029

Candidate for Bachelor of Science in Mechanical Engineering and Computer Science | GPA: 3.607

Relevant Coursework: Robot Dynamics & Control, Algorithms & Data, Statics, Materials Science, Calculus 3, Discrete Structures, Thermodynamics, Differential Equations, Physics 1, Program and Design Fundamentals 1

Activities: Northeastern Mars Rover Team

SKILLS & INTERESTS

Programming: Python, C++, MicroPython

Software: CAD (SolidWorks, Onshape), Git, VS Code, AutoCAD, MATLAB, Arduino IDE

Prototyping & Hardware: 3D printing, Soldering, ESP32, Raspberry Pi Pico, Arduino

Interests: Art(Drawing/Sketching), Rock Climbing

EXPERIENCE

EDR| STEM Instructor| Brockton, MA |

Summer 2026

- Taught hands-on STEM courses in Arduino programming, 3D printing, mathematics, and science to 60 students in grades 6-8.
- Guided students through designing and troubleshooting 20 engineering projects while introducing core computer science and design concepts
- Developed and adapted technical lessons to make engineering, math, and programming concepts accessible to students at varying experience levels

Northeastern Mars Rover Team | Software & Mechanical Engineer |

2025 – Present

- Developed an A* path-planning logic as a prototype to improve upon the current frontier search Algorithm.
 - Designed and fabricated custom mounts and retention components for rover electronics, including USB hubs, a PCB, and Ethernet cabling
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PROJECTS

5-DOF Robotic Arm | SolidWorks, 3D Printing, MATLAB, ESP32 |

August 2026 – Present

- Designing and fabricating a 5-DOF robotic arm with custom geared joints, carbon-fiber linkages, and servo/DC motor actuation
- Redesigned a previous prototype with lighter structural components, improved joint transmissions, additional degrees of freedom, and a more robust mechanical architecture
- Developing joint-control and kinematics software for arm simulation and eventual ESP32-based hardware control

Carnival Crane Machine | Arduino, SolidWorks, 3D Printing |

Spring 2025

- Designed and fabricated an electromechanical crane game with motorized gantry motion and joystick control for a First-Year Engineering final project
- Developed an object-oriented control architecture in MicroPython using a state machine to manage game logic and system behavior on a Raspberry Pi Pico
- Reconfigured the system into an Arduino-controlled ball-transfer game following an electronics failure on demo day, delivering a functional system for the final demonstration